Unit 9 - Computer Networks
Assignment 14: Managing a LAN Task 6

## Introduction

**Task description -** *You are a junior consultant for T5 Alliance. The work you have done so far for Competition Bikes Ltd has been very well received. As they have no IT support staff of their own, they are interested in outsourcing the ongoing management to your company. First, however, you must prove your knowledge of network management.*

**Task 6**

## (U9:D2) - Evaluate typical services available from a network operating system Directory Service

**Task description -** *Research Microsoft Active Directory and highlight the different services that it provides, stating why each service is important.*

*Describe each of the following services;*

- *Active Directory Domain Services*
- *Active Directory Certificate Services*
- *Active Directory Federation Services*
- *Active Directory Lightweight Directory*
- *Active Directory Rights Management*

*State why each service is important.*

Introduction

In this documentation, I will be researching the Microsoft Active Directory and explaining the services that is provided by the Active Directory, stating why they are important. To elaborate on Microsoft Active Directory, or simply known as AD, it is essentially a Windows OS directory service in which it facilitates on working with interconnected, complex and different network resources in a collective manner.

Essentially, Active Directory provides a common interface for organizing and maintaining information related to resources connected to a variety of network directories. These directories may be systems-based, such as the Windows operating system, application-specific or network resources, such as printers. To put it simply, Active Directory serves as a single data storage service for quick data access to all users and controls access for users based on the directory's security policy.

Active Directory Domain Services

Active Directory Domain Services, or simply AD DS, is essentially a database and a set of services that enables managers/administrators to centrally administer a Microsoft Windows network. To elaborate, an active directory domain service stores data for objects, which is known as a domain.

These domains consist of a grouping of categories, such as users, PCs, printers and services within a server. While also providing support to find and work with the said objects. The main features of directory domain services is location transparency, object data, rich query and high availability. Meaning that AD DS can be used as the centre, or one of the centres, of information within a network. With the ability to store and administrate the data that is held within the domain services, to keep them secure and manageable.

Due to this, Active Directory Domain Services is potentially the biggest prolific role in the Active Directory suite. The reason being is that Active Directory is both the directory information source and the service that makes the information available and usable. For example, information about servers, users, clients, network devices, applications and (specifically) email servers. Due to the fact there's a relatively close tie to active directory in the exchange, found within the Microsoft portfolio.

To elaborate, this information is both stored and provided to each of the mentioned categories in the Active Directory Domain Services. Aspects such as manageability, security and interoperability are all provided to the network manager in a centralized, easily managed method. Making it much easier to carry these aspects out than without this ability, due to the technology used.

In retrospect, the reason that Active Directory Domain Services is important, is essentially due to the fact it makes a scalable, secure and manageable infrastructure for user and resource management, all in a single service.

Overall, the reason why Active Directory Domain Services is used, is because it gives network managers the ability to manage basically as many objects, users, computers, printers and even shares as they can possibly store within it. Given that the hardware which supports this is available for the network manager.

In a centralised manner, all from a single console, potentially it also allows the manager to essentially delegate that administration out to other network managers/administrators within the building or even the company that they are working in. Making AD DS the foundation for a Windows Server, which uses Domain controllers.

Essentially, the domain services provides a secure, structured and hierarchical method of storing different types of data for the objects within the network. Giving administrators the ability to store and manage information about network resources. As mentioned before, these resources are: servers, clients, users, network applications and other various pieces of information.

It also provides support for directory enabled applications, such as Microsoft Exchange server, usually when there could be a direct connection to the Active Directory by the exchange server. Allowing for centralized management, then the delegation of that centralized management out to other potential business network managers or administrators.

Active Directory Certificate Services

Active Directory Certificate Services, or AD CS, is basically the Microsoft implementation of public key infrastructure, or PKI for short. Public key infrastructure is essentially a set of hardware, software, people, policies and even the procedures needed to create, manage, distribute, use, store and revoke digital certificates.

In essence, Active Directory Certificate Services creates the responsibility of services, end-user entities, users/computers, certificate stores (responsible for repositories and revocation), requests, certificate retrieval and certificate revocation.

In terms of certificates, which is known as digital certificates, is almost directly similar to physical certificates, that are issued by an authority so they implement a level of security. Which can be done by only providing the access of certain users/devices to a server, customers to a trusting paying/banking/account website or allowing both sides of a communication to be protected from being viewed by an anonymous user.

The importance that Active Directory Certificate Services brings to the network, is that it essentially provides customizable services for issuing and managing digital certificates. Meaning that a certificate is essentially an electronic document which contains a variety of data fields, which work off a type of trust model.

When in use, digital certificates are usually deployed via a hierarchy, with the top priority being the root certificate authority. Which can be an internet certificate authority or an external authority. So, when an authority issues a certificate, they will perform a number of checks on the individual purchasing the said certificate, to ensure that they are valid.

Also, when a certificate is used, it can be checked to see which authority issued said certificate. In order for the certificates to be used, the computer must trust the authority that it was issued from, which is usually trusted by default on a majority of operating systems.

However, In order for this level of security to work properly, the user using the certificate has to have the confidence where the certificates are issued. For instance, if a certificate is presented to the computer and it is not trusted, then the computer will generate an error asking if the users want to trust the certificate. With the user being responsible in the decision if they want it to be valid or not.

Overall, Active Directory Certificate Services, or AD CS, is an Active Directory tool that lets administrators customize services in order to issue and manage public key certificates. These certificates are used in order to confirm the true identity of a device or user, to give authorization to encrypt, sign or access data.

Meaning that the importance of having a service to issue these certificates are vital due to the fact they provide secure access to specific areas of the network, or the network in its entirety. Making sure that, if the server is configured as a CA (certification authority), the server will provide the features of management for the regulation, distribution and overall use of these certificates.

Basically, AD CS issues certificates to users and computers for authorisation/legitimacy, manages the lifetime for these said certificates and revokes any certification and vertifies the status on revocation for the security of a company's PKI, or public key infrastructure.

Active Directory Federation Services

Active Directory Federation Services, or AD FS, is basically a software component that facilitates the organisation for the access for all the systems and applications across the network. Giving the ability to share resources out on the network or giving access to the user's resources in another network.

Though, it depends on which side of the network the network manager/administrator happens to be on. The way it does this, is by providing a simplified secured identity federation and web single sign-on, also known as SSO, capabilities. With federation services being essential, since it gives the manager/administrator the ability to make it a consistent and smooth process.

AD FS is important because it enables the creation of authorisation, provides access to applications and even provides a single sign-on between two different directories for web based applications.

Single sign-on, or SSO, is essentially provided by the AD FS to authenticate other users outside of a company's network. This is so it can enable the users to actually have a simple sign-on available to sign in to the business, which increases the security and authorisation o a company's web-based applications.

Also, AD FS has the capability of sharing information over web services. Meaning that the exchange of information, with the management federated identity, are compatible with various security products with the support of web service paradigm.

Essentially, AD FS follows this web service specification for passive clients, such as browsers, giving the ability for environments that don't use a Windows identity model to centralise with environments which do have a Windows identity model

However, it is not required to have a partner user account management when using AD FS. The reason being that a partner's IP sends claims that essentially indicates the user's identity, groups and other data with their attributes. Meaning that the company will no longer need to change, reset or revoke the credentials for the partner's users.

Another aspect of AD FS is the ability of claim mapping. That is basically the ability for each partner to be understood and mapped out within the AD FS trust policy. Which is mostly used for the exchange between the different partners within the federation.

Additionally, there is the ability to manage a centralised federated partner. Which is just the management that is performed, while using the Microsoft management console within the AD FS, to all the partners that have been federated.

Overall, Active Directory Federation Services (AD FS) is a feature of the Windows Server operating system, that extends end users' single sign-on, or SSO, access to applications and systems outside the corporate firewall.Meaning that users only have to login once to access an application via the web, with the application using the credentials of the manager/administrator in the company.

This is done to basically authenticate and authorise the manager/administrator into said application, in another network or even another organisation. Due to the expandible paradigm AD FS has for claim augmentation. Such as adding or modifying claims processing to use a company's strict validity and policies.

Active Directory Lightweight Directory

Active Directory Lightweight Directory Services is essentially identical to the Active Directory Domain Services, with the difference being that it is using it's template form. Basically, Lightweight Directory Services is an empty shell of what Directory Domain Services are.

Essentially, it is both the directory information source and the services that makes it available and usable, which is the same definition used in Active Directory Domain Services. With the same options as Active Directory Domain Services, with the exception of some of the key abilities that Active Directory Domain Services provides.

AD LD is important for a network because it gives the ability to store, replicate, access information (which is the same information which could be tracked in Active Directory Domain Services), names, locations and application information, which is specific to said application.

This information can be accessed and used by using Active Directory Lightweight Directory Services. Furthermore, Lightweight Directory Services provides the flexible support for directory enabled applications, without the dependencies and domain related restriction of Active Directory Domain Services.

This due to the fact that AD DS is a management tool for an organisation, hence it comes with the overhead of that requirement due to the fact it is a management tool. Which can provide a majority of benefits in terms of management, in terms of functionality, operational and overall benefits that the service has over AD DS.

The functional benefits that AD LDS bring is that it essentially uses the same technology as AD DS, in terms of directory service. Meaning that there is a somewhat similar framework for the network operating system of AD DS with the services of applications that AD LDS has. Therefore increasing the reusability of both the design and overall coding.

Furthermore, AD LDS has the possibility of increasing the scalability of directory services. This is generally done by separating the network operating system services from the rest of the application services. AD LDS also has the possibility for authentication and access control, only through the Windows security principals however since it doesn't have a feature for security on it's own.

The benefits that AD LDS has over AD DS is that it doesn't take into account the overhead of all types of domains, nor does it require the deployment of said domains or even the domain controllers. Also, multiple instances can run correctly on a single AD LDS installation, only if they are configured to a specific application however.

Additionally, due to the structure of AD LDS in comparison to AD DS, the configuration set of AD LDS has a separate config schema. Meaning that it is completely independent to the AD DS schema, hence why they don't fall under the same category.

Overall, AD LDS is essentially used for providing directory services for directory enables applications without the incurring of the overhead of domains and forests, meaning that there is no requirement for a single schema throughout a forest. Basically, the manager/administrator can customise AD LSD as much as possibly desired.

Hence the definition of AD LDS, which is a Lightweight Directory Access Protocol (LDAP) directory service designed for use with directory-enabled applications. A directory-enabled application is one that uses a directory, as opposed to a database or flat file, for its data store.

Basically, Active Directory Lightweight Directory Services is important for a network because it is a method of information storage, similar to AD DS. However, it is different to AD DS in structure, due to the fact that it is a structured information store based on the structure of AD DS. Meaning that it has been modified from AD DS to be much more lightweight, hence the naming.

Active Directory Rights Management

Active Directory Rights Management Services, or AD RMS, is basically an information protection technology, by using both server side and client side components.

Essentially, it gives the manager/administrator/author the ability to set what could be done with the said data after it's creation. Either through the server side component, usually made up of a variety of web services run on a Microsoft server, or through a client side component, usually run on a client or server operating system.

To elaborate, it works with applications to safeguard the digital information within the network, through the encryption and decryption of content. Therefore allowing the author to create content such as a Word document or even an email, allowing to protect said word document/email using those AD RMS aware applications, such as MS Word or MS Outlook.

Active Directory Rights Management is important due to the fact that sensitive information within an Active Directory environment is a significant responsibility since it can cause a plethora of problems when obtained by the wrong person. Which AD RMS is implemented within a server role which aim to stop this from happening, by having its own set of tools to help with security technology to manage the rights on an organisation's data.

Essentially, it allows individuals and administrators to specify access permissions to documents, workbooks and even presentations for example. However, it is not only restricted to just these three pieces of information, for example emails could also be used.

For instance, the manager/admin/author can type up the email, assigning the application to protect the said document so it would restrict access from an unauthorised user to manipulate the document, print it, copy it and even has the ability forward it onto another email recipient. Meaning that it can be expanded out to other forms of data, prevent sensitive information from being printed, forwarded or even copied by unauthorised users.

Then, after giving the said protection, it would keep that document secure no matter where it is being sent to. If the recipient of that information has access to the Rights Management Service infrastructure, either through a public or private connection, they would be able to verify that the document is indeed accessible to the recipient. However, they would still be restricted with the manipulation of the actual document.

Also, it gives access and usage restrictions, which are enforced no matter where the information is located. For instance, if a word document is created, it would be protected with the Rights Management Service then emailed to another user outside the network or even the company.

Essentially giving the recipient the restrictions of that document, however only if the user is given the ability to access the Rights Management Services infrastructure, through specific means, in the first place.

Overall, Active Directory Rights Management Services, or AD RMS (originally known as Windows RMS, the name was changed AD RMS in Windows Server 2008), is a Microsoft Windows security tool that provides persistent data protection by enforcing data access policies for documents to be protected with AD RMS, the application the document is associated with must be RMS-aware.

The way that this is done is by having information, in the form of data, protected by encryption by the AD RMS and stored in the AD RMS server. Which a user would only be able to read/access the file if they have the encryption key in order to decrypt the said information.

Meaning that, if a user has the privilege of accessing the information, they will instantly be able to retrieve the key from the server and open the encrypted file. Which can be carried out with the document being stored elsewhere, however the user would need to contact the AD RMS server in order to retrieve the key to decrypt the information.

Basically, AD RMS gives the network the ability to add security to valuable information within a server, preventing unauthorised access to the said information by encryption. Only giving the method of decrypting the encrypted information via a key.

Conclusion

In conclusion, Active directory is essentially a type of directory service utilised in networking. This particular service is constructed into a variety of server operating systems, such as Microsoft Windows, and benefits both server side and client side operations.

Meaning that the main role and purpose of this directory service is that it is in control for authenticating and empowering of every consumer and node in the network. Which is done by placing security policies for every node within the network itself.

Furthermore, Active Directory is the prime position and situation for network security and management. For instance, if an employee logs into a computer within the organisation’s network, the Active directory will know and recognise whether this student is a common consumer or the system management.

Although it is a directory service that was established by Microsoft, it is also an appliance of internet standard directory and appointing protocols that utilises a database motor for transactional assistance and helps a range of requested operating interface standards.

Moreover, Active Directory is sectioned into single or conjoined domains. A domain is a cluster of users, computers and many various resources that are obtained and kept under control with a specific set of regulations, in other words it is a safeguard border.

Every domain contains a name, individual databases and policies that are suitable to every resource involved with that domain. It is also looked after by a server computer known as a domain controller. A domain controller conducts every account and password of the user for a domain and also sustains the policies and gives the authentication to the consumers of the domain.

Furthermore, the Active Directory has 5 key main services, which are: Domain Services, Certificate Services, Federation Services, Lightweight Directory Services and Rights Management Services.

The Domain Services, or AD DS, is basically a service with a database paradigm in order to store information about certain objects, such as users and devices, within in a network in hierarchical structure. This is so the information is stored available to be accessed for all the different users within the network.

The Certificate Services, or AD CS, is basically a service which allows the administrators or managers to build a public key infrastructure, also providing key cryptography to the public, digital certificates and digital signature capabilities to the organisation. This is so, rather than going on a third party Certificate Authority to get PKI certificates, the users could instead handle this within their own software.

The Federation Services, or AD DS, is basically a simple but secure identity federation via SSO ,or web single sign-on, capabilities. Essentially enabling the users to authenticate using their own credentials and the access to data within the cloud. This is so it has some sort of infrastructure to allow for this access in the first place, whether it is onsite or on the cloud.

The Lightweight Directory Services, or AD LDS, is basically a lightweight version of AD DS, meaning that it is a modified version of the database paradigm that is found with the AD DS to store data. It has the same purpose of AD DS due to the similarities, however AD LDS doesn’t require the creation of domains or domain controllers.

The Rights Management Services, or AD DS, is basically a service which focuses on the rights that each user within the network have. Which is done through encryption and some form of selection functionality denial. In order to limit access to certain documents such as Microsoft

Word documents and web pages, with only the authorised users being able to access and use them

Overall, there are many important uses and advantages of each services found within Active Directory. As it will allow an organisation's consumers to log in to the network with the ability to access data, and have some sort of hierarchy of authority, usually by utilising usernames and passwords that are accustomed somewhere else.

Essentially, it is simple to operate, supervise, conduct and portion information or other resources, such as the organisation's files and other documents. Providing data integrity and security, through any means necessary provided with all these services, due to the fact that information stored within these networks are extremely sensitive to the organisation and need sophisticated containment, accessibility and protection.